

Guanine Exchange Factors Show Existence of Transient Pockets on Rac1 Surface

Elvis A. F. Martis,[†] Morgane Rousselle,[‡] Vincent Sauzeau,[‡] and Stéphane

Téletchéa^{*,†}

[†]*US2B, Nantes Université, CNRS, UMR 6286, F-44000 Nantes, FRANCE*

[‡]*UMR_S 1087 / UMR_C 6291 l'Unité de Recherche de l'Institut du Thorax, F-44000 Nantes, FRANCE*

E-mail: stephane.teletchea@univ-nantes.fr

Supplementary Information

Table S1: Details of stages of Heating using NVT Ensemble

Stage	Temperature range (K)	Time (ns)
Stage 1	0 to 100	0 to 2.5
Stage 2	100 to 200	2.5 to 5
Stage 3	200 to 300	5 to 7.5
Stage 4	300 to 300	7.5 to 10

Table S2: Parameters used for pocket identification in POVME2

Parameter	Value
pocket detection resolution	4.0 Å
pocket measuring resolution	1.0 Å
clashing cutoff	3.5 Å
number of neighbors	4
number of spheres	5
sphere padding	5.0 Å

Table S3: Results of Pocket Detection using Fpocket

Pocket	Fpocket Score	Druggability Score	Hydrophobicity Score
Site P1	0.13	0.001	46.38
Site P2	0.051	0.002	10.89
Site P3	0.076	0.02	27.5
Site P4	0.11	0.005	37.33

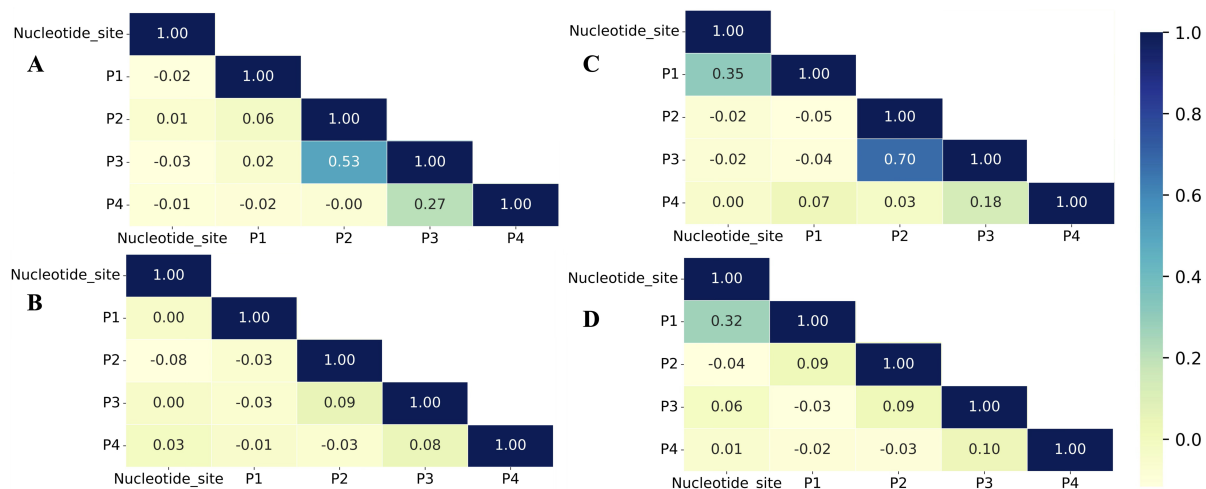


Figure S1: Pearson's Correlation for the volume fluctuations among different pockets and the effect of nucleotide binding. (A) GDP-bound Rac1; (B) GTP-bound Rac1; (C) Nucleotide-free state created by deleting GDP from Rac1; (D) Nucleotide-free state created by deleting GTP from Rac1.

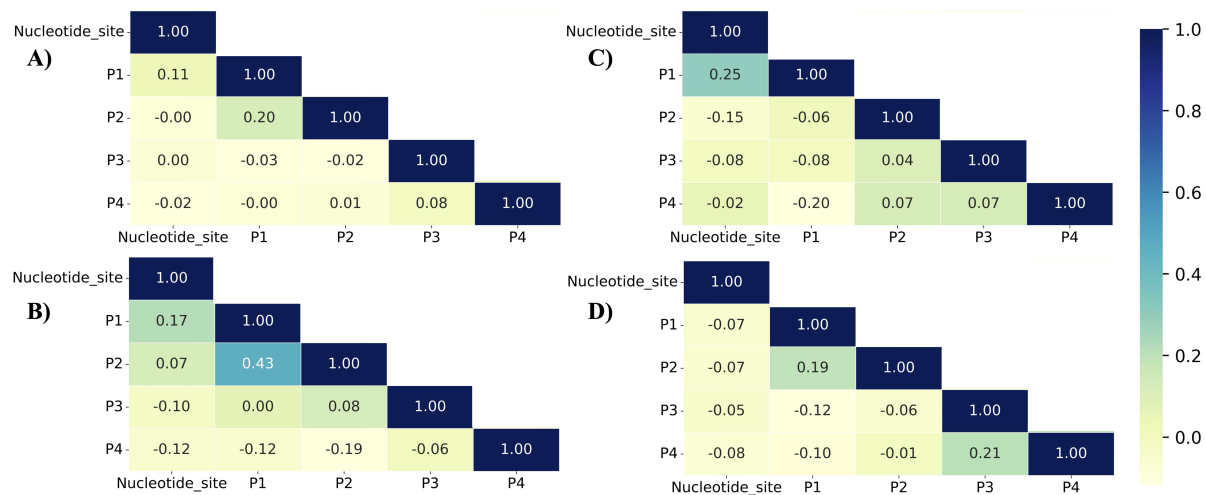


Figure S2: Pearson's Correlation for the volume fluctuations among different pockets and the effect of GEF binding. (A) PREX1-bound Rac1; (B) TIAM1-bound Rac1; (C) VAV1-bound Rac1; (D) DOCK2-bound Rac1. Rac1 in all GEF-bound systems is in the nucleotide-free state.

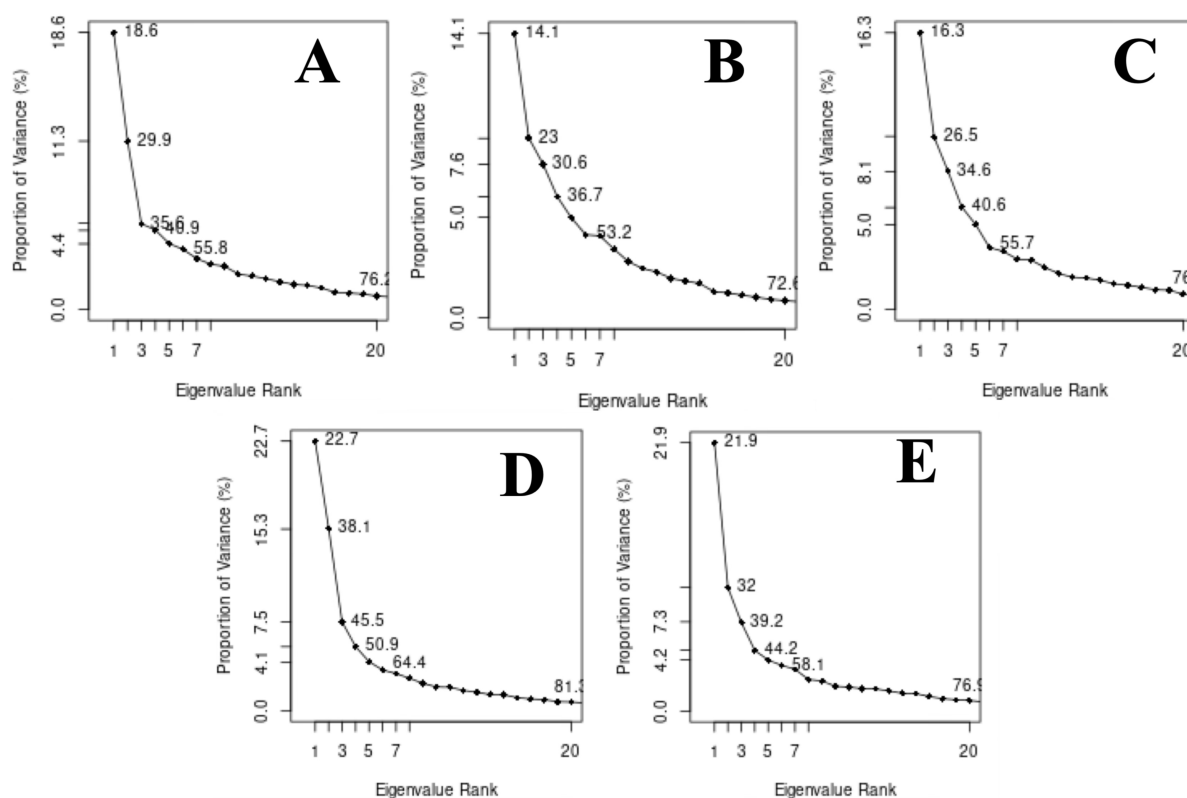


Figure S3: Scree plot to compute the cumulative variance explained by principal components. (A) *apo*-Rac1 in TIP3P water box; (B) GTP-bound Rac1 in TIP3P water box; (C) *apo*-Rac1 in TIP3P water+isopropanol box; (D) *apo*-Rac1 in TIP3P water+phenol box; (E) *apo*-Rac1 in TIP3P water+toluene box.

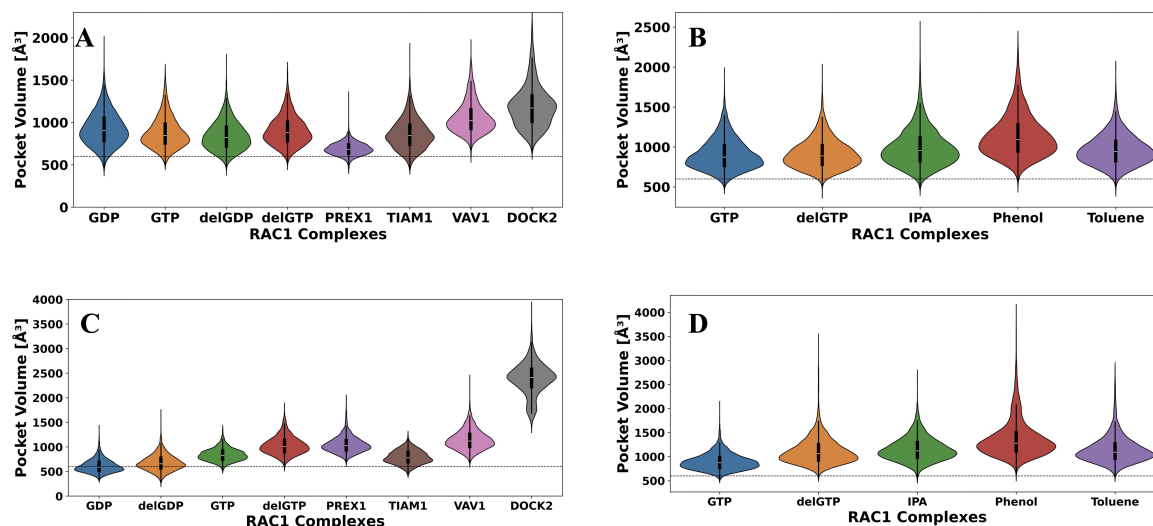


Figure S4: Analysis of pocket volume changes by combining pockets P1 with P2, and P3 with P4. **(A)** Combined pocket volume analysis for site P1+P2 ; **(B)** Combined pocket volume analysis for site P1+P2 in co-solvent; **(C)** Combined pocket volume analysis for site P3+P4; **(D)** Combined pocket volume analysis for site P3+P4 in co-solvent. The threshold value, beyond which a pocket may be considered amenable for small molecule binding, was set according to the following main text references^{36–38} and indicated by the dashed line.

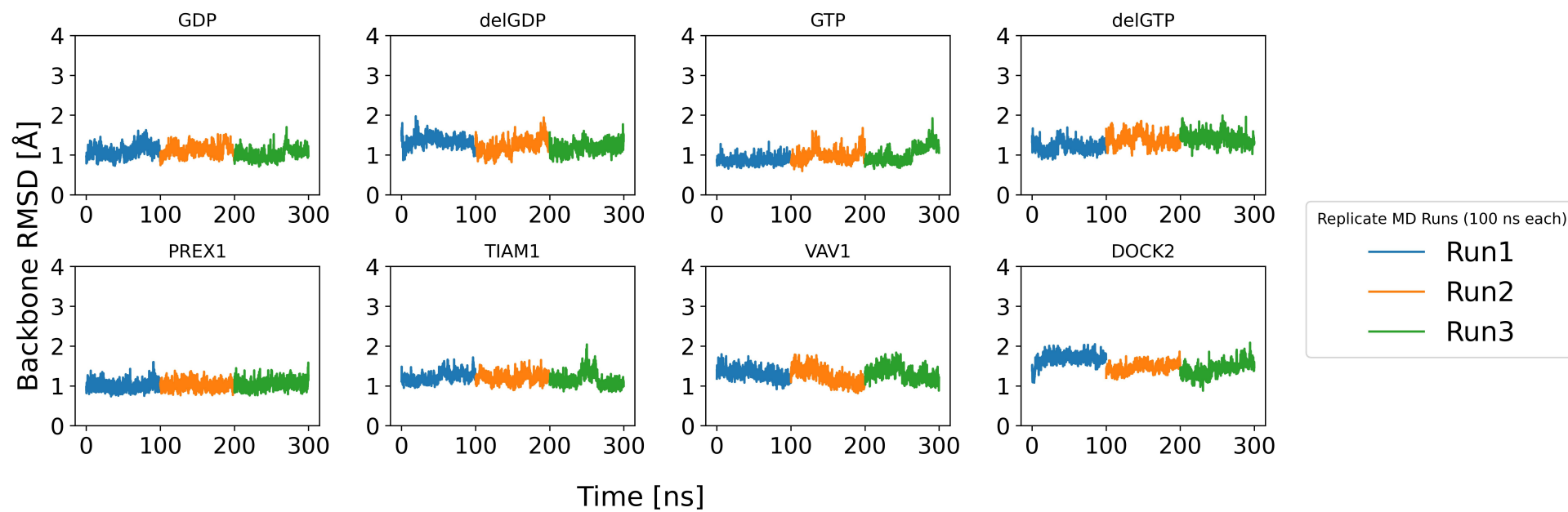


Figure S5: Time-evolution of RMSD of backbone atoms from Full Rac1 protein.

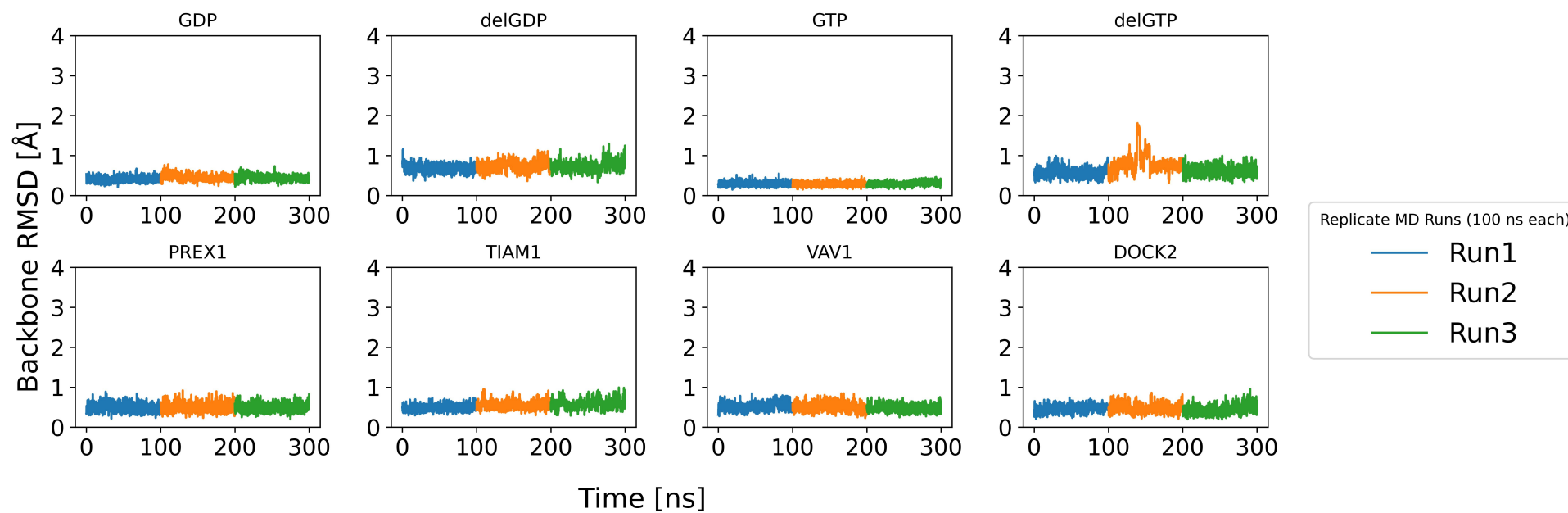


Figure S6: Time-evolution of RMSD of backbone atoms from the P-loop region.

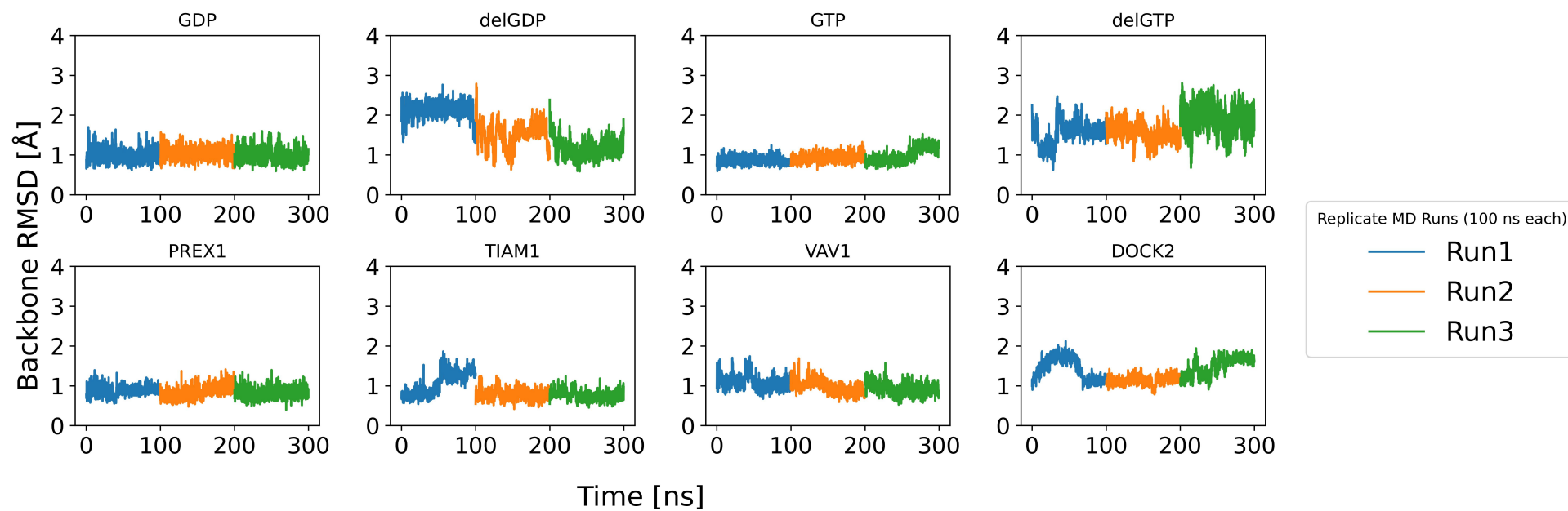


Figure S7: Time-evolution of RMSD of backbone atoms from the switch I region.

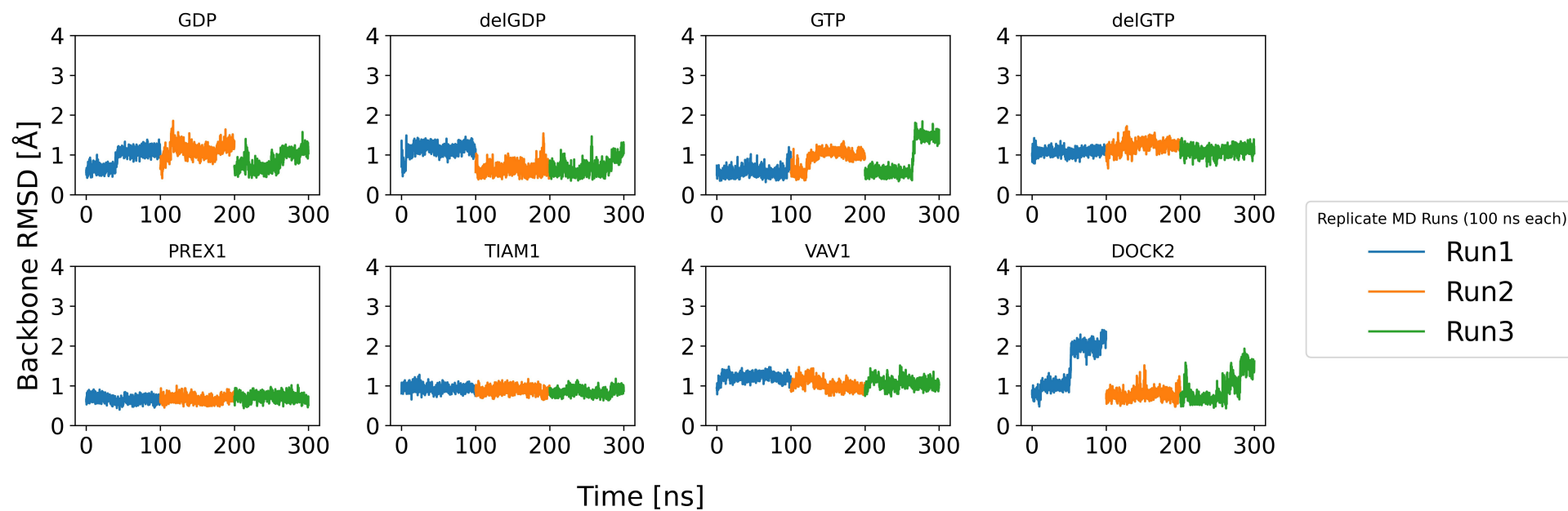


Figure S8: Time-evolution of RMSD of backbone atoms from the switch II region.

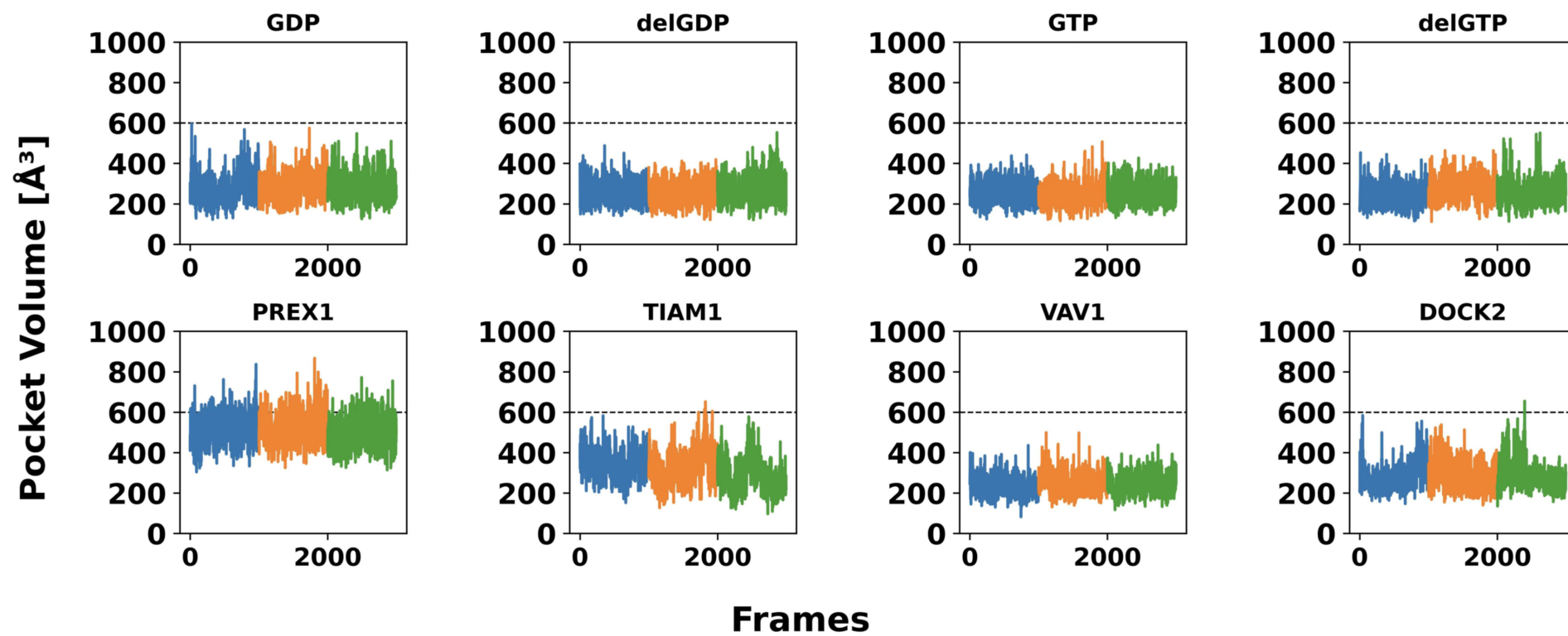


Figure S9: Time-evolution of pocket volume for Site P1.

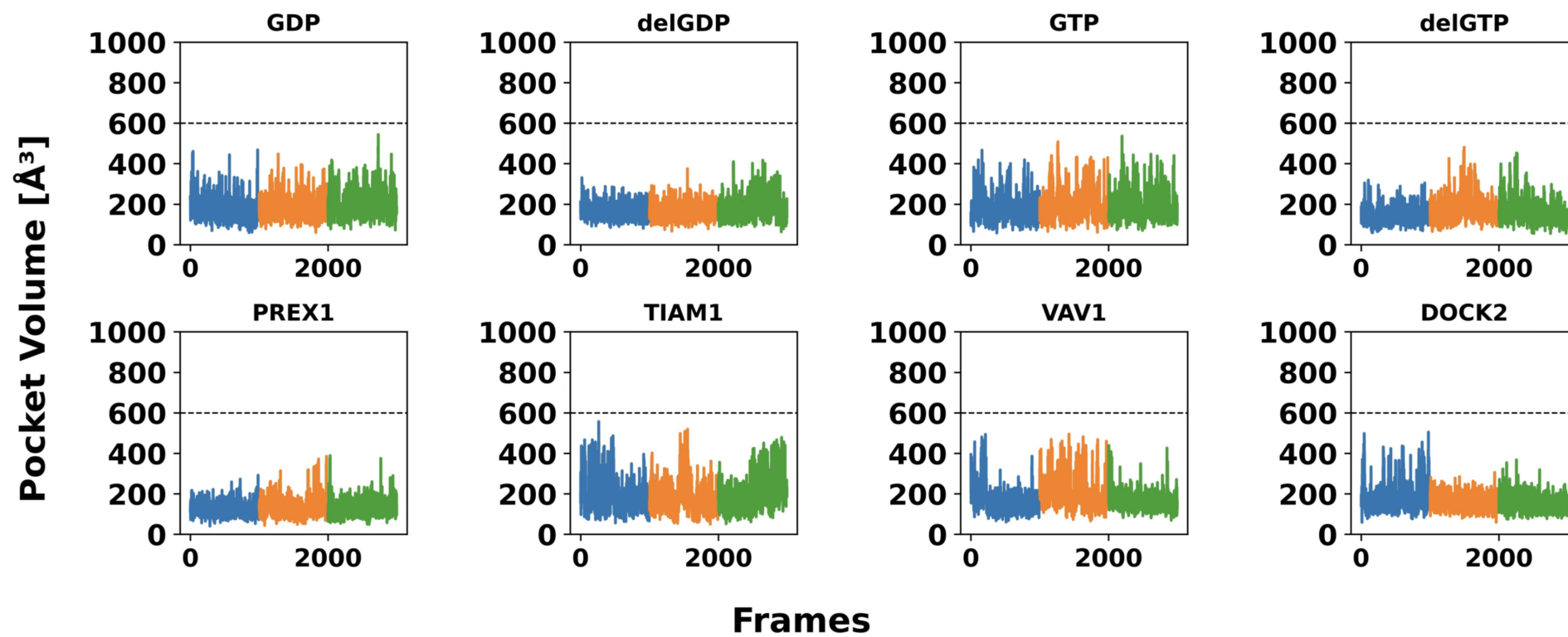


Figure S10: Time-evolution of pocket volume for Site P2.

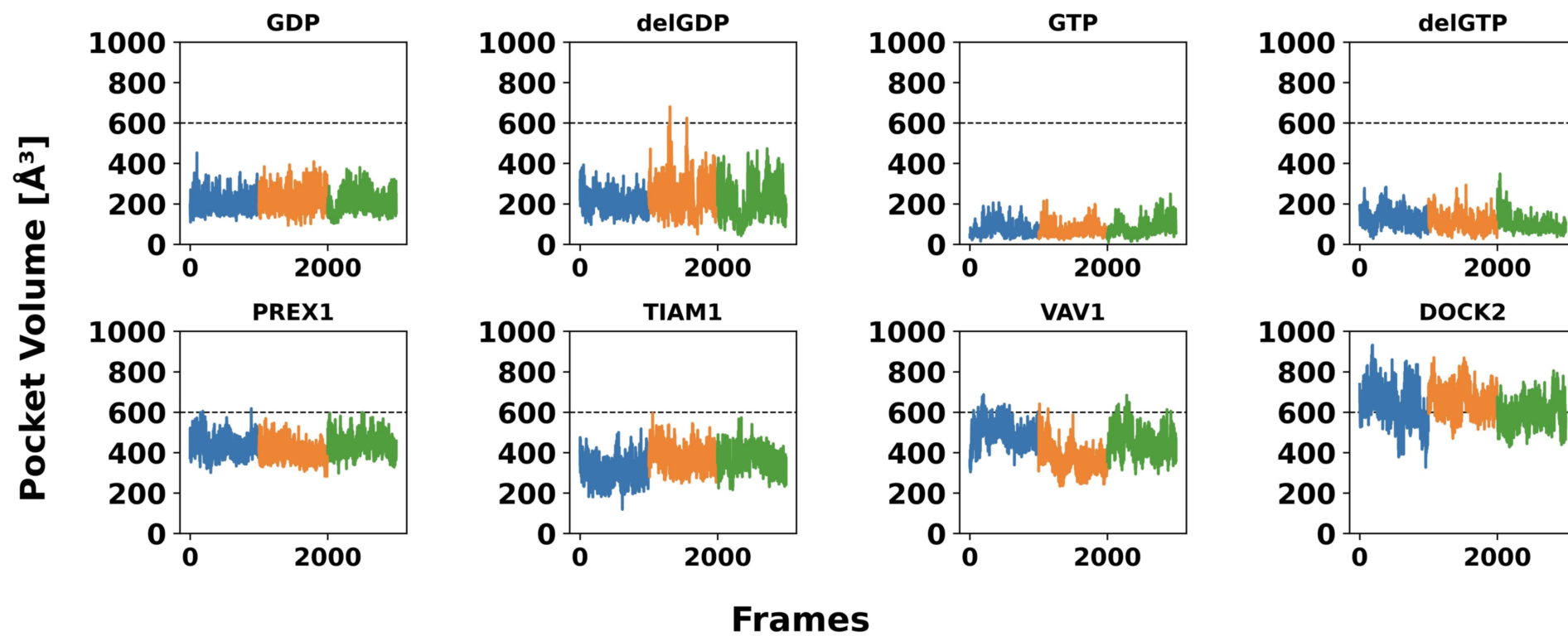


Figure S11: Time-evolution of pocket volume for Site P3.

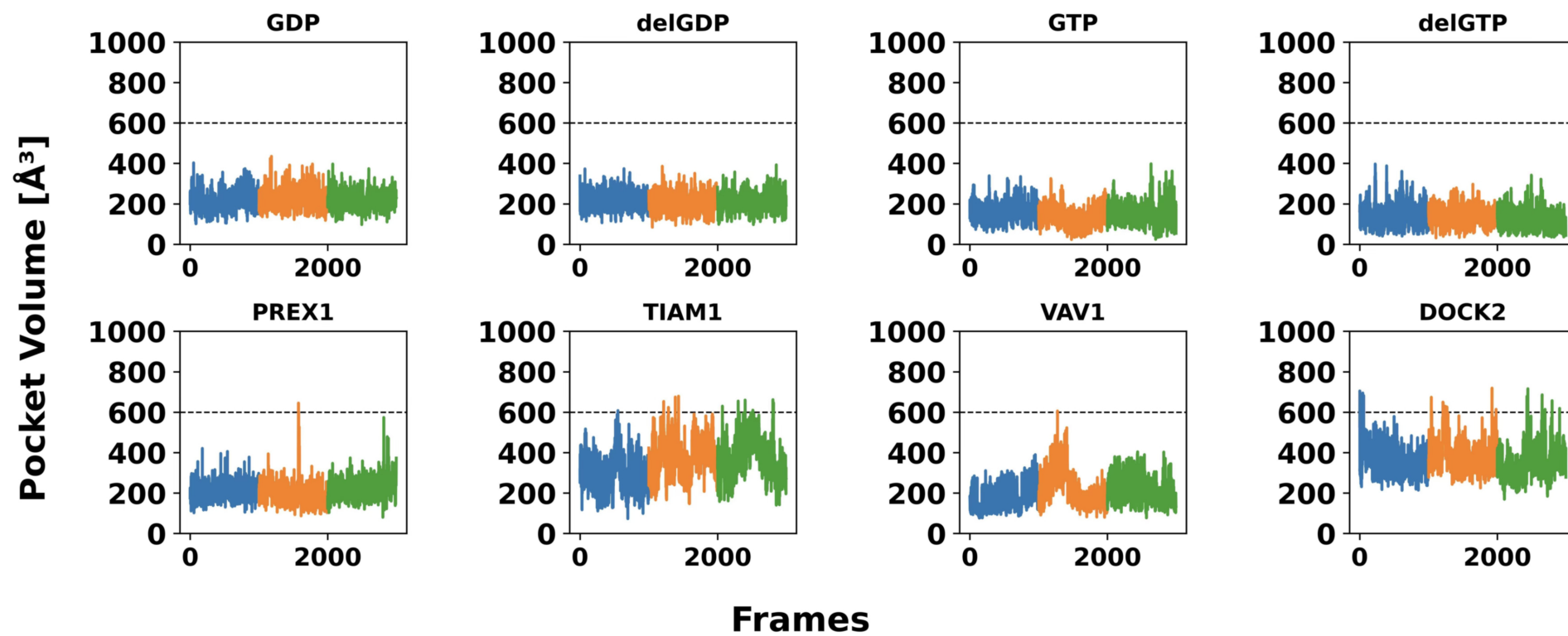


Figure S12: Time-evolution of pocket volume for Site P4.

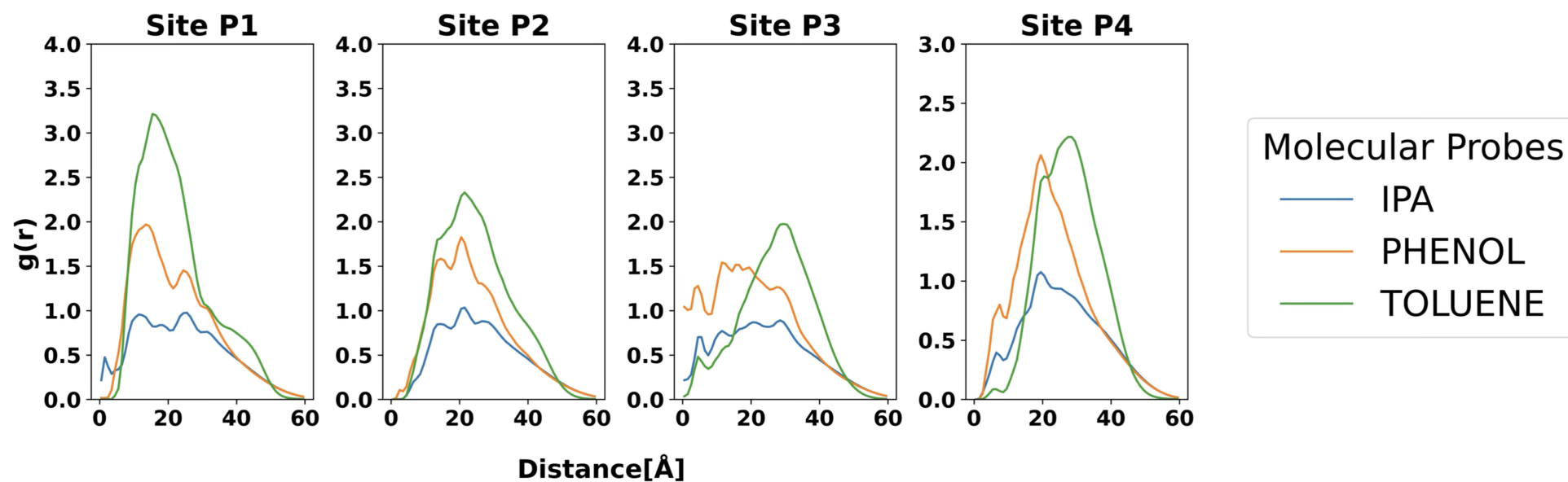


Figure S13: Radial Distribution Function of the Molecular Probes computed for each of the new pockets identified in this work.

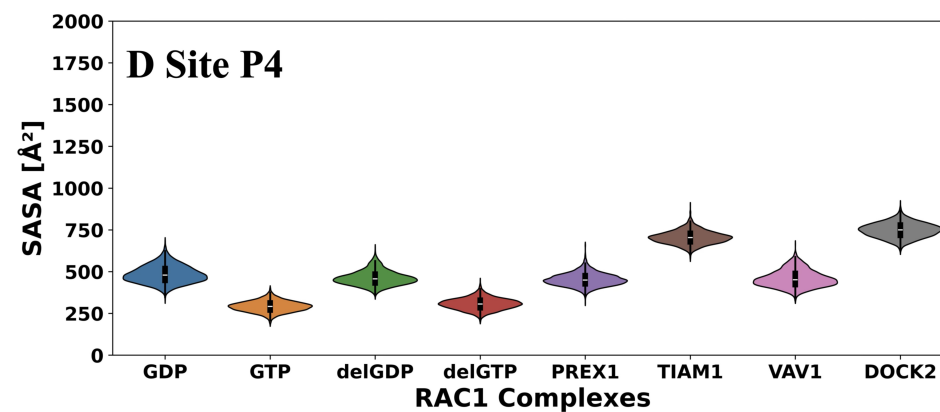
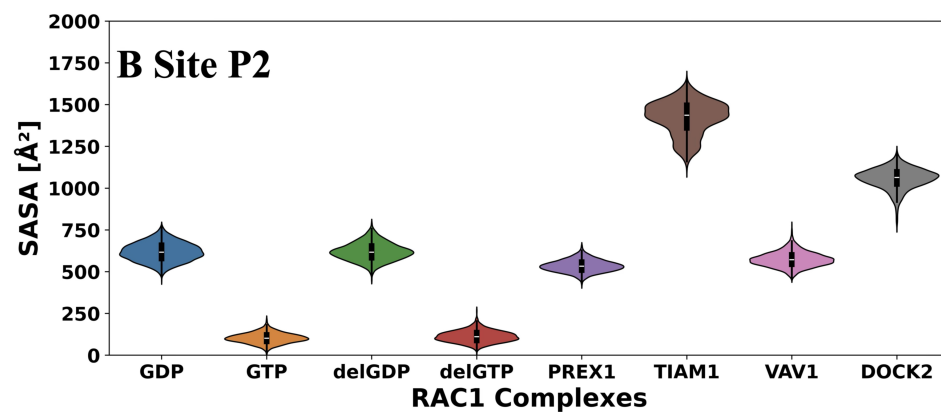
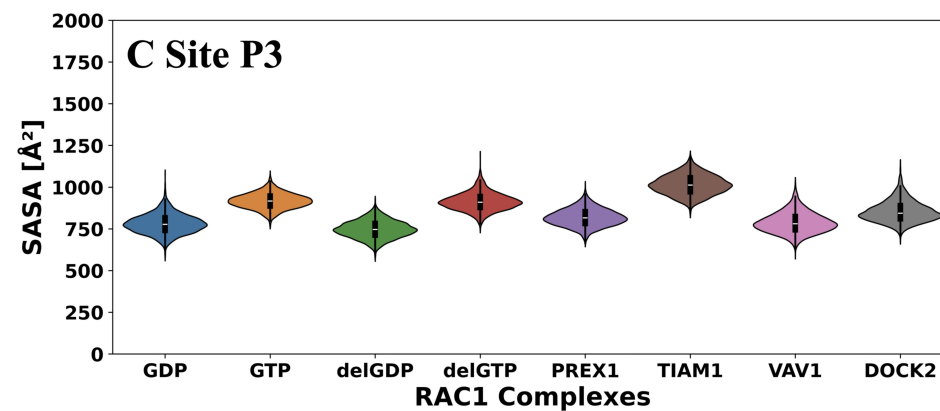
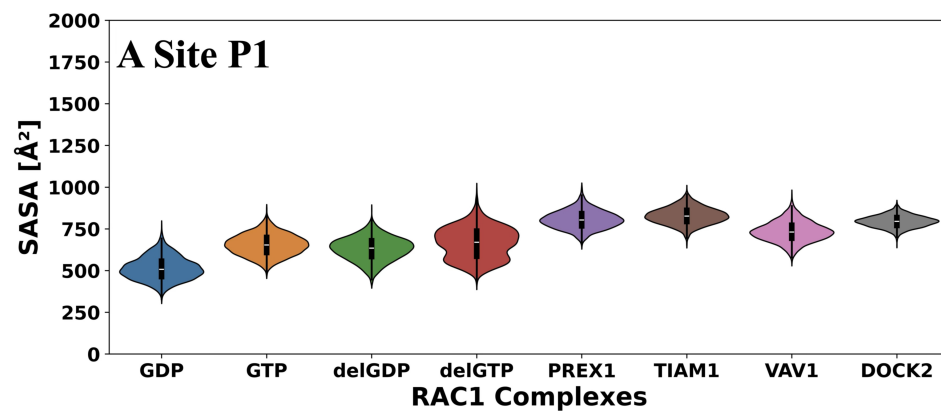


Figure S14: Analysis of the Solvent-Accessible Surface Area using a 1.4 $\text{\AA}$ probe across all the newly identified site to understand the effect of nucleotide and GEF binding

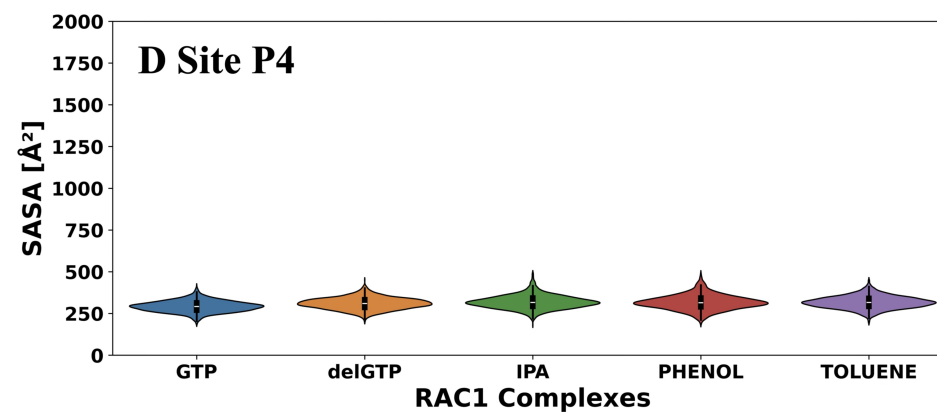
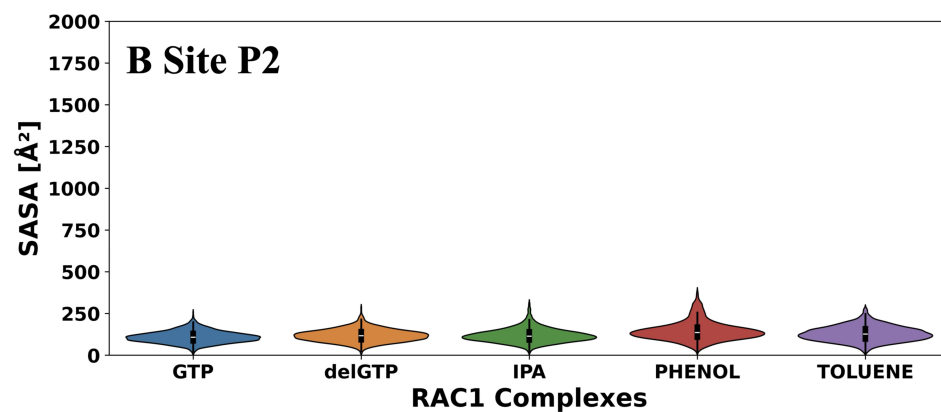
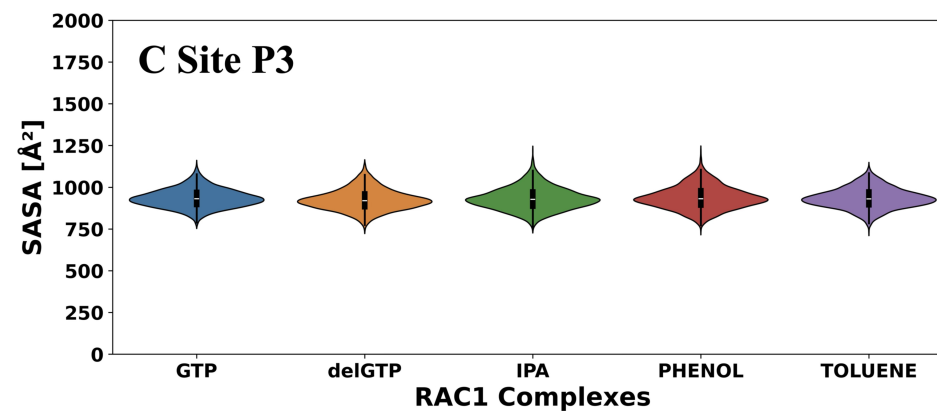
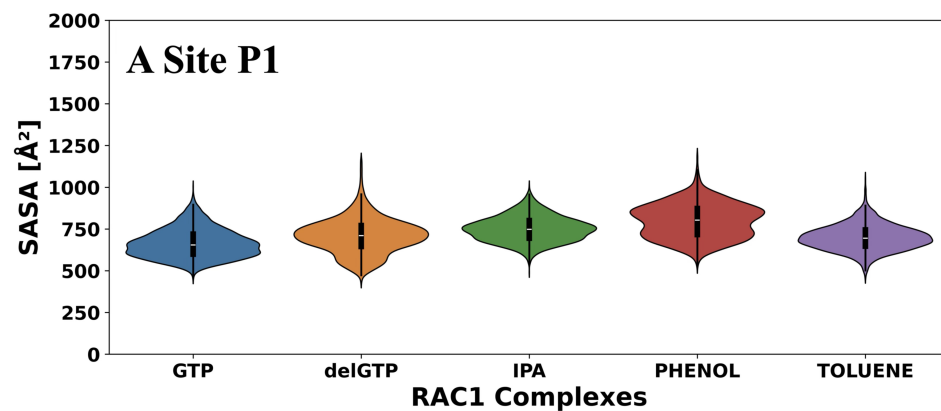


Figure S15: Analysis of the Solvent-Accessible Surface Area using a 1.4 Å probe across all the newly identified site to understand the effect of Molecular probes